

Leveraging Temporal Distribution Shifts for Rice Disease Detection

Anonymous Author(s)
Submitted for blind review

Abstract—Rice diseases threaten agricultural yield and food security in Bangladesh. Most automated detection studies train and test on images from a single time period, so temporal generalization goes unmeasured. We evaluate deep learning models trained on Bangladeshi rice field images from 2021 to 2025 and tested only on held-out 2026 data. Four architectures were compared under full and partial transfer learning: ResNet50, EfficientNet-B0, DenseNet121, and DINOv2. Two claims were specified before testing and assessed with seed-matched paired tests, assumption checks, Holm correction, effect sizes, and confidence intervals. The shift is visible in the images themselves, which lose 52.06% of their structural similarity across the partition boundary against a 9.29% within-partition control. Over 10 seeds, future-year macro-F1 fell below validation macro-F1 in all eight model/strategy configurations (Holm-adjusted $p < 3 \times 10^{-13}$). The mean generalization gap was 43.91 macro-F1 points (95% CI: 39.42–48.39) and mean retention only 48.28%. Fine-tuning strategy mattered as well, but its effect depended on the architecture. Full fine-tuning was better for the three CNNs and partial fine-tuning for DINOv2, a heterogeneity that was itself significant (Friedman $\chi^2 = 28.08$, $p = 3.49 \times 10^{-6}$) and echoed in how far each backbone’s attention was reorganized. DINOv2 with partial fine-tuning gave the best future-year macro-F1 even though it did not attain the best validation score.

Index Terms—rice disease detection, deep learning, temporal generalization, distribution shift, convolutional neural networks, vision transformers, transfer learning, statistical hypothesis testing

I. INTRODUCTION

Rice disease threatens food security for over 160 million people in Bangladesh and costs smallholder farmers much of their income [1]. Rice blast, brown spot, and rice tungro can destroy up to 70% of a yield in a severe outbreak [2], yet diagnosis still depends on expert agronomists who are largely unavailable to rural farmers. CNNs and vision transformers now make automated recognition from field photographs practical at scale [3]–[6].

The difficulty is that most classifiers are trained and tested on images from one season. Lighting, growth stage, camera equipment, and disease progression vary from year to year, and a single-season benchmark captures none of it [7]–[10]. High same-year accuracy may therefore vanish as soon as the model meets the following season, leaving a gap between laboratory numbers and real usability [11]–[15]. We train on 2021–2025 images and evaluate only on a held-out 2026 partition, which measures how well ResNet50 [16], EfficientNet-B0 [17], DenseNet121 [18], and DINOv2 [19] survive inter-year distributional shift [20]–[22]. Fig. 1 shows the framework.

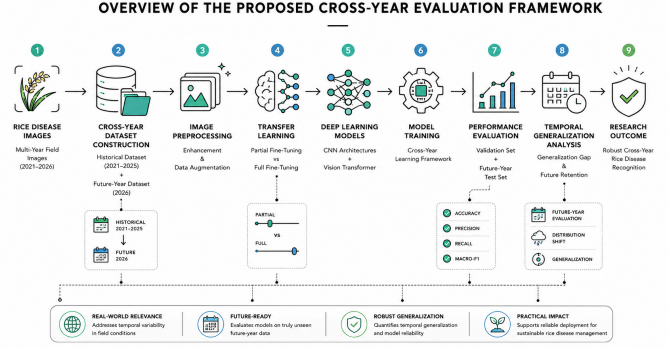


Fig. 1. Overview of the proposed cross-year evaluation framework: models are trained on historical (2021–2025) rice disease images and evaluated on a held-out future-year (2026) test set.

We built the study around two claims fixed before any testing. The *primary claim* is that temporal shift significantly reduces recognition performance, tested by comparing same-period validation against future-year test performance. The *secondary claim* is that fine-tuning strategy significantly affects cross-year performance, and that the better strategy depends on the architecture; we test it within each architecture and then across them. Every configuration uses the same 10 seeds, so comparisons are paired at the seed level and the data decide each effect’s direction.

Our contributions are as follows:

- We propose a temporal evaluation framework for rice disease detection using chronologically separated training (2021–2025) and future-year testing (2026) to assess temporal distribution shifts [10], [12].
- We systematically compare four pretrained deep learning architectures under full and partial transfer learning, quantifying both input-level distribution shift and cross-year performance [20]–[22].
- To the best of our knowledge, this is the first study to investigate temporal cross-year generalization for rice disease detection using multi-year Bangladeshi field datasets.
- We evaluate temporal degradation and fine-tuning effects using a seed-matched statistical framework with paired hypothesis tests, effect sizes, confidence intervals, and multiplicity correction, showing that optimal fine-tuning is architecture-dependent [14], [15].

II. LITERATURE REVIEW

A. Deep Learning for Plant Disease Detection

Early CNN work established that deep networks classify plant disease accurately from leaf images [3]–[5]. ResNet [16], DenseNet [18], and EfficientNet [17] became the usual baselines because pretrained ImageNet weights existed for them [20], [23], [24]. ViT [21] and DINOv2 add self-supervised pretraining, which helps when labelled agricultural data are scarce [6].

B. Rice Disease Detection and Research Gap

Rice disease has been attacked with CNNs, transformers, ensembles, and detectors such as YOLO [7], [25]. Work built on controlled datasets like PlantVillage [26] reports high accuracy that does not survive real fields, where backgrounds are cluttered and image quality uneven [9], [15]. Islam et al. [8] found the same across geography: models trained on international datasets transfer poorly to Bangladeshi images.

Most of this work optimizes accuracy on randomly partitioned data, so training and test images share a temporal distribution and the reported numbers can overstate what deployment would give. Only a few agricultural studies look at year-over-year domain shift [11]–[13], and Bangladesh-specific multi-year evaluations are rarer still. A methodological gap runs alongside it. Cross-year degradation is usually reported from single runs, without repetition or significance testing, so a reader cannot tell a systematic effect from run-to-run noise. Fine-tuning strategies are likewise presented as one universally better choice, not as an effect that might depend on the backbone. We therefore evaluate CNN and transformer models on chronologically separated data with a seed-matched protocol fixed in advance.

III. DATA CURATION

A. Dataset Collection

We combined ten publicly available Bangladesh-focused collections from Kaggle and Mendeley Data [27]–[36], listed in Table IX (Appendix B). Ten independent repositories capture more variation in field conditions, imaging devices, seasons, and locations than any single benchmark [3], [9], [15]. We reviewed all images by hand and discarded duplicates, low-quality and synthetically modified samples, and cases we could not label confidently. Four classes had enough multi-year representation to keep: Rice Blast, Brown Spot, Rice Tungro, and Healthy.

B. Dataset Structure and Temporal Splitting

Table I gives the year-wise distribution. Most public Bangladeshi datasets lack verified acquisition dates, so we used publication year as a proxy. Coverage is uneven: 2021 has no Healthy images and 2022–2023 no Tungro, so those years contribute only to the classes they cover. We handled the imbalance with class-weighted loss and augmentation.

Random train–test splits inflate performance, because training and test images then share a temporal distribution [4], [5],

TABLE I
YEAR-WISE IMAGE DISTRIBUTION ACROSS THE DATASET.

Year	Raw Images	Used Images	Split Role
2021	606	606	Training / Validation
2022	493	493	Training / Validation
2023	729	729	Training / Validation
2024	4,397	4,397	Training / Validation
2025	2,458	2,458	Training / Validation
2021–2025 Total	8,683	8,683	Train: 7,380 / Val: 1,303
2026	7,823	4,000	Test (1,000/class)
Grand Total	16,506	12,683	—

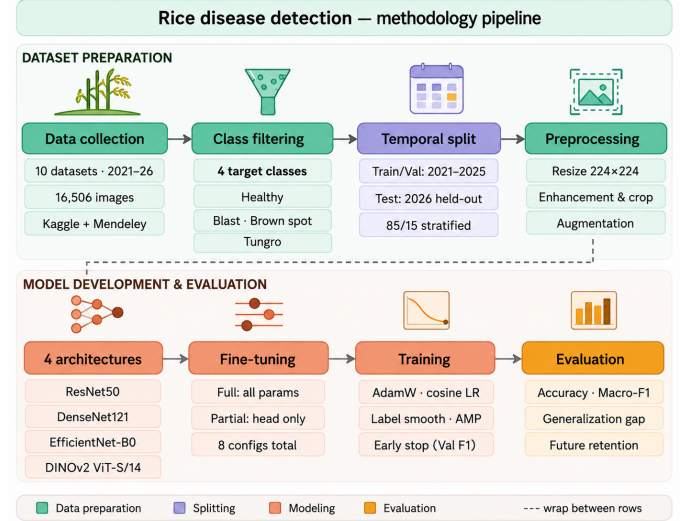


Fig. 2. Overview of the methodology: preprocessing pipeline, model architectures, and full/partial fine-tuning strategies evaluated across historical (2021–2025) and future-year (2026) partitions.

[23]. We used a strict chronological split instead, simulating deployment: a model trained only on historical data must classify unseen images from a later season. This split is the central design of the study [10], [11], [13]. Images from 2021 to 2025 formed the training and validation pool, divided 85/15 with stratification. No 2026 data entered training, tuning, or model selection. For the future-year evaluation we drew 1,000 images per class from 2026 by stratified sampling, giving a balanced test set of 4,000 images.

IV. METHODOLOGY

Chronological partitioning, four architectures under two fine-tuning strategies, and 10 seeds per configuration together allow both study questions to be answered with seed-matched paired tests (Section IV-E). Fig. 2 summarizes the pipeline.

A. Preprocessing and Augmentation

Every image passed through the same pipeline: EXIF orientation correction, RGB conversion, edge-preserving bilateral denoising, CLAHE contrast enhancement on the L-channel in LAB space, a center crop that keeps the leaf region, and resizing to 224×224 . Training added mild augmentation (random flip, rotation, color jitter, affine), while validation and test images received none, so their evaluation is deterministic.

We normalized with ImageNet statistics. Appendix A lists the full parameters.

B. Architectures, Fine-Tuning, and Training

We evaluated ResNet50, DenseNet121, EfficientNet-B0, and DINOv2 ViT-S/14, spanning the CNN and Vision Transformer families. Each started from ImageNet-pretrained or self-supervised weights with its classification layer replaced for the four-class task. The two strategies were *full* fine-tuning, which updates all parameters, and *partial* fine-tuning, which freezes the backbone and trains only the head.

Training used AdamW with cosine annealing and class-weighted cross-entropy with label smoothing; Appendix A lists the hyperparameters. Early stopping tracked validation macro-F1 and kept the best checkpoint. Each architecture/strategy pair was trained with 10 seeds, for 80 runs under identical partitions, preprocessing, and evaluation settings. Macro-F1 is our primary metric because it weights every class equally. We also report accuracy, the generalization gap (validation minus future-year test performance), and future retention (the percentage of validation performance kept). Experiments used PyTorch, torchvision, and timm.

C. Quantifying the Input-Level Shift

We wanted to know whether the partitions differ as images, not only in the scores they produce. Two full-reference metrics were computed on the preprocessed 224×224 images of Section IV-A, so resolution, orientation, and contrast normalization cannot account for the outcome. For images x and y ,

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \quad (1)$$

where μ , σ^2 , and σ_{xy} are means, variances, and covariance, and C_1, C_2 are stabilizing constants. Peak signal-to-noise ratio is

$$\text{PSNR}(x, y) = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}(x, y)} \right), \quad (2)$$

with $\text{MSE}(x, y) = \frac{1}{HWC} \sum_{h,w,c} (x_{hwc} - y_{hwc})^2$ over height H , width W , and C channels. Because images were scaled to $[0, 1]$, $\text{MAX}_I = 1$. Higher values of either metric mean greater similarity. SSIM responds to structure and contrast, PSNR to raw intensity, so the pair separates a change in scene structure from a change in exposure and colour.

We compared three sets of pairs under one sampling and pairing procedure, which makes their values comparable: Train–Train and Test–Test, drawn inside the historical and future-year partitions, and Train–Test, drawn across them. To express the change without reference to either metric’s units, we report the relative reduction of a similarity measure μ from the within-historical baseline,

$$\text{reduction} = \frac{\mu_{\text{Train–Train}} - \mu_{\text{Train–Test}}}{\mu_{\text{Train–Train}}} \times 100\%, \quad (3)$$

applied to SSIM and then to PSNR. Train–Train is the reference because it describes the data the models actually

saw. Putting $\mu_{\text{Test–Test}}$ in place of $\mu_{\text{Train–Test}}$ gives a within-partition control, which separates the reduction due to the temporal boundary from the reduction due to ordinary variation between two different images. Both metrics depend on how pairs are formed, so we read them only as a relative comparison across the three sets and report them descriptively.

D. Attention Analysis

To see *where* each model looks, and whether fine-tuning moves it, we computed Grad-CAM saliency maps for every architecture under both strategies. All models saw the same 20 held-out 2026 images, five per class. We compared the partial and full maps three ways: their Pearson correlation, where 1.0 means identical focus, the intersection-over-union of their most activated regions, and the centroid shift as a fraction of the image diagonal. Values were averaged over the 20 images inside each seed, then compared across architectures with the paired tests described next.

E. Statistical Analysis

Every configuration was trained with the same 10 seeds, so all comparisons are paired at the seed level. Pairing removes seed-to-seed variation and raises power. We fixed two confirmatory claims in advance. For the *primary claim*, that temporal shift reduces performance, each condition was tested as $H_0: \mu_{\text{Val}} = \mu_{\text{Test}}$ against the one-sided $H_1: \mu_{\text{Val}} > \mu_{\text{Test}}$. For the *secondary claim*, that fine-tuning strategy affects cross-year performance, each architecture was tested as $H_0: \mu_{\text{Full}} = \mu_{\text{Partial}}$ against the two-sided $H_1: \mu_{\text{Full}} \neq \mu_{\text{Partial}}$ on future-year macro-F1. We chose a two-sided alternative deliberately. We did not assume that either strategy always wins, and the data decide which one does for which architecture.

Shapiro–Wilk assessed normality of the paired differences at $\alpha = 0.05$. Differences consistent with normality went to the paired t -test, the rest to the exact Wilcoxon signed-rank test. We report both either way, so a reader can see how far a conclusion depends on that choice. We report the mean paired difference with its 95% confidence interval, the test statistic, and Cohen’s d_z , the mean difference divided by the standard deviation of the differences. For rank-based tests we add the Wilcoxon z and the matched-pairs rank-biserial correlation. Holm’s step-down procedure controlled the family-wise error rate inside each family of comparisons.

Two further analyses support these claims. To summarize the primary claim at study level, we repeated the paired test with the eight condition means as units of analysis, avoiding the pseudoreplication of pooling 80 non-independent runs. To test architecture dependence head-on instead of inferring it from four separate tests, we formed each architecture’s seed-matched full-minus-partial difference vector and compared the vectors with a Friedman test and Kendall’s W , followed by paired comparisons of the difference-of-differences. We used SciPy throughout.

V. RESULTS

All results cover the 10 seed-matched runs and are given as mean $\pm$ standard deviation. Future-year test macro-F1 is the

primary metric. Table II collects the significance tests for both claims.

A. Input-Level Evidence of Temporal Shift

We first looked for the shift in the image data alone, without a classifier, using (1) and (2) as described in Section IV-C. Pairs drawn inside a partition resembled one another. Train–Train averaged 0.872 ± 0.061 SSIM and 31.84 ± 3.42 dB PSNR, Test–Test 0.791 ± 0.078 and 28.57 ± 3.16 dB. Pairs drawn across the partitions did not: 0.418 ± 0.113 SSIM and 18.92 ± 4.08 dB, a loss of 12.92 dB. Applying (3), crossing the temporal boundary costs 52.06% of structural similarity and 40.58% of pixel-level fidelity.

The within-partition control settles the interpretation. Applied to Test–Test, the same expression gives reductions of just 9.29% and 10.27%, so the cross-partition reduction is more than five times the control for SSIM and four times it for PSNR. Variation between two different images accounts for only a small part of the dissimilarity. The temporal boundary accounts for the rest.

B. Cross-Year Performance

Table III summarizes the eight conditions. Future-year performance fell below validation performance in every one. DINOv2 partial reached the best future-year macro-F1 (47.11 ± 1.32 ; 95% CI: 46.17–48.06) and the best retention, yet ranked only fifth of eight on validation macro-F1. ResNet50 full took the opposite position, best on validation and worst on the gap. Section V-C tests these gaps. Across the 80 runs, validation macro-F1 tracked test macro-F1 positively but loosely (Pearson $r = 0.641$, $p = 1.56 \times 10^{-10}$; Spearman $\rho = 0.621$), and across the eight condition means the association was not significant at all ($r = 0.666$, $p = 0.0714$). Ranking models on same-period validation would not have found the most temporally stable configuration.

C. Primary Claim: Temporal Shift Reduces Performance

Shapiro–Wilk found the paired differences compatible with normality in all eight conditions ($p = 0.185$ – 0.991), so the paired t -test served as the primary test in each, with the exact Wilcoxon signed-rank test as a distribution-free check. Future-year macro-F1 was significantly lower in every condition (all Holm-adjusted $p < 3 \times 10^{-13}$). Wilcoxon agreed everywhere, returning $p = 0.00098$ with $z = 2.803$ in all eight. That value is the smallest obtainable at $n = 10$, and it appears because all 10 seed-level differences carried the same sign, which is also why the parametric and rank-based tests agree. Per-condition gaps ran from 34.63 points for DINOv2 partial to 49.20 points for ResNet50 full, with Cohen’s d_z between 19.17 and 72.09. The drop is more than an order of magnitude larger than the seed-level variability around it.

We then treated the eight condition means as the units of analysis, so that each configuration contributes one observation and the runs are not pseudoreplicated. The mean gap was 43.91 macro-F1 points (95% CI: 39.42–48.39; Shapiro–Wilk $p = 0.137$; $t(7) = 23.14$, $p = 3.56 \times 10^{-8}$; $d_z = 8.18$), and

mean future retention was $48.28 \pm 4.46\%$. Accuracy gave the same picture, with a gap of 42.13 points ($t(7) = 23.31$, $p = 3.39 \times 10^{-8}$), so the result is not an artifact of macro-averaging.

We reject H_0 . Under every architecture and strategy we tried, models kept less than half of their validation macro-F1 when moved to future-year samples. Read together with the input-level shift of Section V-A, this is the central finding of the study.

D. Secondary Claim: Architecture-Dependent Fine-Tuning

The two strategies differed significantly for all four architectures (all Holm-adjusted $p \leq 0.0020$), but they did not differ in one direction. Full fine-tuning won for EfficientNet-B0 (+11.32 points, 95% CI: 10.35–12.29; $d_z = 8.38$), DenseNet121 (+8.46, 6.47–10.45; $d_z = 3.04$), and ResNet50 (+4.59, 3.45–5.72; $d_z = 2.90$). Partial fine-tuning won for DINOv2, which gave up 9.08 points when fully fine-tuned (95% CI: -10.07 to -8.08 ; $d_z = -6.52$). Shapiro–Wilk flagged non-normal differences for ResNet50 alone ($p = 0.044$); Wilcoxon was used there and matched the parametric result.

Four separate tests cannot show that the effect differs *between* architectures, so we tested that directly, treating each architecture’s seed-matched full-minus-partial difference as an effect vector. A Friedman test found the effect is not homogeneous ($\chi^2 = 28.08$, $p = 3.49 \times 10^{-6}$, Kendall’s $W = 0.936$), and the DINOv2 effect separated from every CNN: ResNet50 (difference of differences -13.66 , 95% CI: -15.06 to -12.27 ; $p = 3.69 \times 10^{-9}$), DenseNet121 (-17.54 ; $p = 1.78 \times 10^{-8}$), and EfficientNet-B0 (-20.40 ; $p = 4.89 \times 10^{-11}$). We reject H_0 . Fine-tuning strategy significantly affects cross-year performance, and which strategy to prefer depends on the architecture.

The Grad-CAM analysis of Section IV-D suggests why the two families behave so differently. In the CNNs, moving from partial to full fine-tuning left much of the saliency structure in place. Correlation between the partial and full maps was 0.460 ± 0.027 for DenseNet121, 0.421 ± 0.032 for ResNet50, and 0.308 ± 0.031 for EfficientNet-B0. In DINOv2 the two maps were unrelated (0.018 ± 0.056), and the overlap of their most activated regions was smaller to match (IoU = 0.178 ± 0.020 , against 0.335–0.402 for the CNNs). Paired tests separated DINOv2 from every CNN on both measures: correlation (differences -0.289 to -0.442 ; all Holm-adjusted $p \leq 5.5 \times 10^{-8}$) and IoU (differences -0.157 to -0.224 ; all Holm-adjusted $p \leq 1.8 \times 10^{-8}$). The effect was again heterogeneous across architectures ($\chi^2 = 28.08$, $p = 3.49 \times 10^{-6}$, $W = 0.936$). Full fine-tuning thus refines what a CNN already attends to but reorganizes what DINOv2 attends to. That is consistent with the reading that updating all parameters adapts an ImageNet-pretrained CNN to the domain, while for DINOv2 it displaces a general-purpose self-supervised representation that would otherwise carry across years. The evidence is associative. It shows that attention moves much further for DINOv2, not that this movement causes the loss in accuracy.

TABLE II

SIGNIFICANCE TESTS FOR THE TWO CLAIMS. PRIMARY: $H_0: \mu_{\text{Val}} = \mu_{\text{Test}}$ VERSUS $H_1: \mu_{\text{Val}} > \mu_{\text{Test}}$ (ONE-SIDED). SECONDARY: $H_0: \mu_{\text{Full}} = \mu_{\text{Partial}}$ VERSUS $H_1: \mu_{\text{Full}} \neq \mu_{\text{Partial}}$ (TWO-SIDED). ALL TESTS ARE PAIRED ON SEED-MATCHED RUNS; THE TEST IS CHOSEN BY A SHAPIRO–WILK CHECK ON THE PAIRED DIFFERENCES ($\alpha = 0.05$), AND p -VALUES ARE HOLM-ADJUSTED WITHIN EACH FAMILY.

Comparison	n	Mean diff.	95% CI	Shapiro–Wilk p	Selected test	Holm p	Cohen’s d_z	Decision
<i>Primary claim: validation – future-year test macro-F1</i>								
ResNet50 full	10	49.20	[48.19, 50.22]	0.185	paired t -test	4.50×10^{-15}	34.6	Reject H_0
ResNet50 partial	10	41.59	[41.18, 42.01]	0.740	paired t -test	1.22×10^{-17}	72.1	Reject H_0
DenseNet121 full	10	47.92	[46.56, 49.28]	0.991	paired t -test	5.84×10^{-14}	25.2	Reject H_0
DenseNet121 partial	10	37.64	[36.37, 38.91]	0.217	paired t -test	1.87×10^{-13}	21.2	Reject H_0
EfficientNet-B0 full	10	47.65	[46.92, 48.39]	0.734	paired t -test	5.40×10^{-16}	46.4	Reject H_0
EfficientNet-B0 partial	10	45.63	[44.93, 46.33]	0.307	paired t -test	5.40×10^{-16}	46.6	Reject H_0
DINOv2 full	10	47.01	[46.12, 47.90]	0.729	paired t -test	2.58×10^{-15}	37.7	Reject H_0
DINOv2 partial	10	34.63	[33.33, 35.92]	0.265	paired t -test	2.28×10^{-13}	19.2	Reject H_0
All conditions (macro-F1)	8	43.91	[39.42, 48.39]	0.137	paired t -test	3.56×10^{-8}	8.18	Reject H_0
All conditions (accuracy)	8	42.13	[37.86, 46.40]	0.150	paired t -test	3.39×10^{-8}	8.24	Reject H_0
<i>Secondary claim: full – partial future-year test macro-F1</i>								
ResNet50	10	+4.59	[3.45, 5.72]	0.044	Wilcoxon	0.0020	+2.90	Reject H_0 (favors Full)
DenseNet121	10	+8.46	[6.47, 10.45]	0.708	paired t -test	1.00×10^{-5}	+3.04	Reject H_0 (favors Full)
EfficientNet-B0	10	+11.32	[10.35, 12.29]	0.552	paired t -test	3.00×10^{-9}	+8.38	Reject H_0 (favors Full)
DINOv2	10	-9.08	[-10.07, -8.08]	0.198	paired t -test	2.07×10^{-8}	-6.52	Reject H_0 (favors Partial)
Architecture $\times$ strategy	10	—	—	—	Friedman	3.49×10^{-6}	$W = 0.936$	Effect is heterogeneous

TABLE III

CROSS-YEAR PERFORMANCE OF THE EIGHT MODEL/FINE-TUNING CONDITIONS OVER 10 SEEDS (MEAN $\pm$ SD). GAP IS THE VALIDATION-TO-TEST MACRO-F1 DIFFERENCE; RETENTION IS THE PERCENTAGE OF VALIDATION MACRO-F1 RETAINED ON THE 2026 TEST SET.

Condition	Val F1	Test F1	Gap	Retention (%)
ResNet50 full	93.02 $\pm$ 0.34	43.82 $\pm$ 1.44	49.20	47.11
ResNet50 partial	80.83 $\pm$ 0.35	39.24 $\pm$ 0.49	41.59	48.54
DenseNet121 full	92.86 $\pm$ 0.25	44.94 $\pm$ 1.85	47.92	48.40
DenseNet121 partial	74.12 $\pm$ 0.58	36.48 $\pm$ 1.61	37.64	49.22
EfficientNet-B0 full	92.30 $\pm$ 0.30	44.64 $\pm$ 1.04	47.65	48.37
EfficientNet-B0 partial	78.96 $\pm$ 0.39	33.32 $\pm$ 0.72	45.63	42.21
DINOv2 full	85.04 $\pm$ 0.99	38.03 $\pm$ 1.03	47.01	44.73
DINOv2 partial	81.74 $\pm$ 0.64	47.11 $\pm$ 1.32	34.63	57.65

E. Condition Ranking and Stability

Over all eight conditions the Friedman test gave $\chi^2 = 64.00$, $p = 2.39 \times 10^{-11}$, $W = 0.914$. All seven planned comparisons between DINOv2 partial and the other conditions survived Holm correction, with advantages from 2.17 points over DenseNet121 full ($p = 0.031$) to 13.79 over EfficientNet-B0 partial ($p = 0.0059$). Its nearest rivals were the fully fine-tuned CNNs, not the partially fine-tuned ones, which is how the architecture dependence of Section V-D shows up at condition level. Seed variability stayed small next to the temporal shift: the coefficient of variation for test macro-F1 ran from 1.26% for ResNet50 partial to 4.41% for DenseNet121 partial, and DINOv2 partial sat at 2.81%. Its advantage does not rest on one lucky seed.

F. Per-Class Performance and Error Structure

Support was balanced at 1,000 images per class, but DINOv2 partial did not treat the classes alike. Healthy leaves came through well ($F1 = 80.61 \pm 1.18$) and rice blast collapsed (14.78 ± 4.12), with brown spot and rice tungro between them (46.41 ± 1.56 and 46.65 ± 2.47). Table VIII in Appendix A covers every condition.

The mean confusion matrix (Fig. 3) shows where the loss occurs. Brown spot absorbed 72.73% of the rice blast samples,

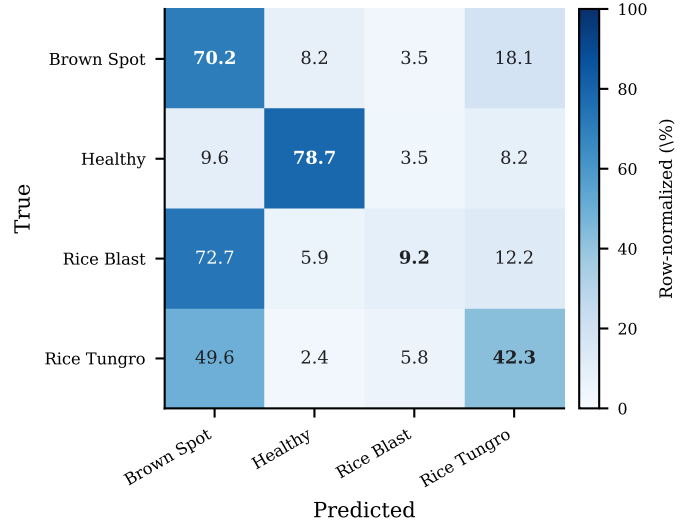


Fig. 3. Row-normalized mean confusion matrix for DINOv2 partial on the 2026 test set, averaged over 10 seeds. Rice blast is predominantly absorbed by brown spot.

which left blast recall at 9.17%, and it pulled in 49.6% of rice tungro as well. The damage is concentrated in one direction and is asymmetric, since brown spot itself was recovered at 70.2%. Under the 2026 shift these models over-predict brown spot and swallow the lesion classes that resemble it.

Fig. 4 makes the same point on two 2026 rice blast samples. In the first, partial fine-tuning settles on the lesion and gets the class right, while full fine-tuning activates on the vegetation around it and answers brown spot. In the second, both strategies look mostly at background and both answer brown spot. Attention that slides off the lesion and onto the canopy goes together with the collapse in blast recall.

G. Key Findings

The shift is present in the images themselves and it cut performance significantly in all eight conditions, with a mean

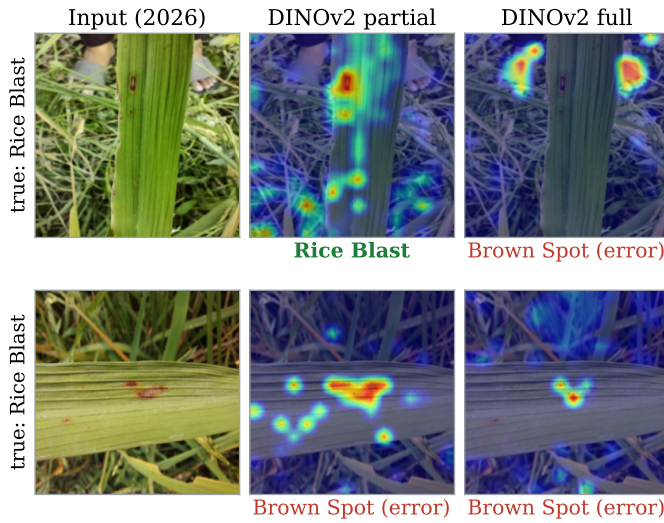


Fig. 4. Grad-CAM maps for DINOv2 on two future-year (2026) rice blast samples. Top: partial fine-tuning localizes the lesion and classifies correctly, while full fine-tuning attends to background and predicts brown spot. Bottom: both attend to background and both predict brown spot, the dominant error of Fig. 3.

gap of 43.91 points and retention of 48.28%. Fine-tuning strategy also mattered, and the strategy to prefer depended on the architecture. DINOv2 partial was the strongest configuration on future-year data despite not having the best validation macro-F1, so validation ranking did not point to the most stable model. Rice blast stayed the hardest class, mostly through confusion with brown spot.

VI. LIMITATIONS

We organize the dataset by the publication year of each source (2021–2026), because most publicly available Bangladeshi rice disease datasets do not report verified acquisition dates, publication year was used as a practical proxy for chronological organization. Consequently, the observed performance differences should be interpreted as cross-year evaluation based on dataset release chronology rather than verified image acquisition chronology.

The statistical results need care as well. Our comparisons are paired across seeds, so the reported p -values and effect sizes say how consistently an effect reproduces across training runs of the same models on the same partitions. They say nothing about sampling variability across datasets, seasons, or regions. The very large d_z values follow from that design, since seed-to-seed variability is small next to the validation-to-test gap. The mean gap and its confidence interval are the better guide to practical magnitude. The conclusions hold for these models on this corpus and should not be treated as estimates that carry over unchanged to new data. The SSIM and PSNR comparisons are descriptive and depend on the pairing procedure.

The study also covers only four disease classes from public datasets. Extending it to more diseases, more crop varieties, and prospectively collected longitudinal images with verified

dates would test temporal generalization more severely than we can here.

VII. CONCLUSION

Automated rice disease recognition has to keep working after deployment, but most deep learning studies report accuracy on randomly partitioned data and never see the temporal shifts that arrive with a new growing season. We built a chronological framework to expose them, training on historical 2021–2025 images and evaluating only on unseen 2026 data, with a 10-seed protocol and seed-matched paired hypothesis testing.

The shift was measurable in the images, which lost 52.06% of their structural similarity across the partition boundary against a 9.29% within-partition control. It cut performance significantly for all four architectures under both fine-tuning strategies, with a mean generalization gap of 43.91 macro-F1 points and less than half of validation performance retained. Strong in-distribution numbers therefore do not carry over to a later year. Fine-tuning strategy mattered too, and which strategy to prefer turned on the architecture: the CNN backbones gained from full fine-tuning while DINOv2 gained from partial fine-tuning, a split mirrored in how far each backbone’s attention was reorganized. DINOv2 with partial fine-tuning gave the best future-year performance, the best retention, and the smallest gap, without ever holding the best validation score, so validation metrics alone did not identify the model that generalized best.

These results argue for temporally separated evaluation in agricultural disease recognition. Beyond high validation accuracy, future work should target temporal shift through better representation learning, domain adaptation, continual learning, and longitudinal field datasets with verified acquisition dates, and should confront the rice blast confusion that still limits future-year performance.

REFERENCES

- [1] Bangladesh Bureau of Statistics, “Yearbook of agricultural statistics 2023,” Tech. Rep., 2023.
- [2] S. Savary *et al.*, “The global burden of pathogens and pests on major food crops,” *Nature Ecology & Evolution*, vol. 3, no. 3, pp. 430–439, 2019.
- [3] J. G. A. Barbedo, “A review on the main challenges in automatic plant disease identification based on visible range images,” *Biosystems Engineering*, vol. 144, pp. 52–60, 2016.
- [4] S. P. Mohanty, D. P. Hughes, and M. Salathe, “Using deep learning for image-based plant disease detection,” *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.
- [5] K. P. Ferentinos, “Deep learning models for plant disease detection and diagnosis,” *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018.
- [6] A. Kamilaris and F. X. Prenafeta-Boldu, “Deep learning in agriculture: A survey,” *Computers and Electronics in Agriculture*, vol. 147, pp. 70–90, 2018.
- [7] J. Chen *et al.*, “A survey of deep learning-based rice disease detection,” *Expert Systems with Applications*, vol. 201, p. 117199, 2022.
- [8] M. T. Islam *et al.*, “Rice disease detection from field images using deep convolutional neural networks: A bangladesh perspective,” *Computers and Electronics in Agriculture*, vol. 198, p. 107064, 2022.
- [9] A. Ramcharan *et al.*, “Deep learning for image-based cassava disease detection,” *Frontiers in Plant Science*, vol. 8, p. 1852, 2017.

- [10] D. Tuia *et al.*, "Domain adaptation for tiled multiresolution land cover classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 50, no. 3, pp. 1114–1125, 2011.
- [11] A. Abdalla *et al.*, "Assessing fusarium oxysporum disease severity in cotton using UAS images and a hybrid domain adaptation deep learning time series model," *Biosystems Engineering*, vol. 240, pp. 1–15, 2024.
- [12] H. Fathipoor *et al.*, "Improved crop disease detection across years via adversarial domain adaptation applied to hyperspectral and thermal imagery," in *Proc. IGARSS 2025*. IEEE, 2025.
- [13] L. Li *et al.*, "Temporal transferability of deep learning-based crop disease detection from UAV imagery," *Remote Sensing*, vol. 13, p. 4567, 2021.
- [14] X. Wu *et al.*, "From laboratory to field: Unsupervised domain adaptation for plant disease recognition in the wild," *Plant Phenomics*, vol. 2023, pp. 1–20, 2023.
- [15] A. Ahmad, A. E. Gamal, and D. Saraswat, "Toward generalization of deep learning-based plant disease identification under controlled and field conditions," *IEEE Access*, vol. 10, pp. 12 345–12 360, 2022.
- [16] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [17] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 97, 2019, pp. 6105–6114.
- [18] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708.
- [19] M. Oquab *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.
- [20] M. T. Ahad, Y. Li, B. Song, and T. Bhuiyan, "Comparison of CNN-based deep learning architectures for rice diseases classification," *Artificial Intelligence in Agriculture*, vol. 7, pp. 56–70, 2023.
- [21] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2021.
- [22] Y. Borhani, J. Khoramdel, and E. Najafi, "A deep learning based approach for automated plant disease classification using vision transformer," *Scientific Reports*, vol. 12, p. 12345, 2022.
- [23] Y. Lu *et al.*, "Identification of rice diseases using deep convolutional neural networks," *Neurocomputing*, vol. 267, pp. 378–384, 2017.
- [24] W. Liang *et al.*, "Rice blast disease recognition using a deep convolutional neural network," *Scientific Reports*, vol. 9, p. 2869, 2019.
- [25] G. Zhou *et al.*, "Recognition of rice diseases and pests in field images based on improved YOLO," *Computers and Electronics in Agriculture*, vol. 169, p. 105174, 2020.
- [26] D. P. Hughes and M. Salathe, "An open access repository of images on plant health: Plantvillage," 2015.
- [27] M. F. Hossain, S. Abujar, S. R. H. Noori, and S. A. Hossain, "Dhan-shomadhan: A dataset of rice leaf disease classification for bangladeshi local rice," [Online]. Available: <https://doi.org/10.17632/znsxdctwt.1>, 2021.
- [28] M. Hasan, S. Khatun, M. A. Raihan, and A. H. Uddin, "Rice leaf bacterial and fungal disease dataset," [Online]. Available: <https://doi.org/10.17632/hx6f852hw4.2>, 2023.
- [29] A. C. Mondal, "Paddy leaf disease," [Online]. Available: <https://doi.org/10.17632/fys884zvyr.1>, 2024.
- [30] S. H. Rifat, T. A. Layes, A. Hasan, and M. U. Mojumdar, "Rice leaf disease and pest dataset overview," [Online]. Available: <https://doi.org/10.17632/vwv3nry3wr.1>, 2024.
- [31] R. Labib, S. Mim, and M. U. Mojumdar, "Rice leaf and crop disease detection dataset," [Online]. Available: <https://doi.org/10.17632/g7tcwvshff.1>, 2024.
- [32] S. A. Rimi and M. J. U. Chowdhury, "RiceLeafBD: A real-field image dataset for rice leaf disease detection and classification in bangladesh," [Online]. Available: <https://doi.org/10.17632/kx9rx8p2mz.1>, 2025.
- [33] M. Tuhin, "Bangladeshi crops disease dataset (BCDD)," [Online]. Available: <https://www.kaggle.com/datasets/musfiqurtuhin/bangladeshi-crops-disease-dataset-bcdd>, 2025.
- [34] I. Ashiqul, "Paddy leaf disease," [Online]. Available: <https://www.kaggle.com/datasets/iashiqul/paddy-leaf-disease>, 2025.
- [35] N. S. Raki, M. A. Bakki, F. Sheikh, M. N. S. Pria, S. Parvin, and M. H. Matin, "Rice disease image dataset," [Online]. Available: <https://doi.org/10.17632/jw7hp6r5gj.1>, 2026.
- [36] M. M. H. Munna *et al.*, "Rice-LeafDiseaseBD: A field-based annotated smartphone image dataset of healthy and diseased rice leaves from bangladesh," [Online]. Available: <https://doi.org/10.17632/86s4jzj2m4.2>, 2026.

TABLE IV
DATASET, PREPROCESSING, AND OPTIMIZATION HYPERPARAMETERS USED IN ALL EXPERIMENTS.

Parameter		Value	Notes
Historical Years	Training	2021–2025	Used for training and validation
Future Test Year		2026	Held-out future-year evaluation
Validation Split		15%	Stratified split of historical data
Number of Classes		4	Healthy, Blast, Brown Spot, Tungro
Input Image Size		224×224	Applied to all models
Image Format		JPEG (Quality = 95)	After preprocessing
Color Space		RGB	EXIF orientation corrected
Image Normalization		ImageNet Mean & SD	Applied to all models
Denoising		Bilateral Filter	Edge-preserving preprocessing
Contrast Enhancement		CLAHE (LAB)	Applied before resizing
Crop Strategy		Center Crop	Preserves leaf region
Training Augmentation		Flip, Rotation, Color Jitter, Affine	Training only
Val./Test Augmentation		None	Deterministic evaluation
Optimizer		AdamW	Used for all experiments
LR (Full Fine-tuning)		1×10^{-4}	All backbone layers trainable
LR (Partial Fine-tuning)		1×10^{-3}	Backbone frozen
Weight Decay		1×10^{-4}	AdamW regularization
Loss Function		Cross-Entropy + class weights + label smoothing	Multi-class classification
Label Smoothing		0.1	Cross-entropy smoothing
LR Scheduler		Cosine Annealing	Applied during training
Maximum Epochs		15	Maximum training epochs
Early Stopping Metric		Validation Macro-F1	Best checkpoint retained
Mixed (AMP)	Precision	Enabled	Used when CUDA available

TABLE V
MODEL-SPECIFIC CONFIGURATION.

Model	Fine-Tuning Modes	Batch Size	Pretrained Weights
ResNet50	Full & Partial	16	ImageNet
DenseNet121	Full & Partial	12	ImageNet
EfficientNet-B0	Full & Partial	24	ImageNet
DINOv2 ViT-S/14	Full & Partial	16	Self-supervised DINOv2

APPENDIX A EXPERIMENTAL CONFIGURATION AND HYPERPARAMETERS

Table IV lists the dataset, preprocessing, and optimization hyperparameters used in all experiments. Table V lists the model-specific configuration, and Table VI lists the overall experimental protocol.

Table VII reports extended descriptive statistics for future-year test macro-F1, including medians, ranges, confidence intervals, and coefficients of variation. Table VIII gives per-class future-year F1 for all eight conditions.

TABLE VI
EXPERIMENTAL PROTOCOL.

Parameter	Value
Evaluation Metric (Primary)	Macro-F1 Score
Secondary Metric	Accuracy
Additional Metrics	Balanced Accuracy, Precision, Recall, Per-class F1
Generalization Metrics	Generalization Gap, Future Retention
Number of Runs	80 (4 architectures $\times$ 2 fine-tuning modes $\times$ 10 seeds)
Framework	PyTorch
Supporting Libraries	torchvision, timm
Hardware Acceleration	CUDA (when available)

TABLE VII
EXTENDED DESCRIPTIVE STATISTICS FOR FUTURE-YEAR TEST MACRO-F1 ACROSS 10 SEEDS.

Condition	Mean	SD	Median	Min	Max	95% CI low	95% CI high	CV (%)
ResNet50 full	43.822	1.443	43.557	42.336	46.672	42.790	44.854	3.292
ResNet50 partial	39.236	0.494	39.407	38.351	39.738	38.883	39.590	1.259
DenseNet121 full	44.941	1.846	44.731	41.754	48.282	43.620	46.261	4.108
DenseNet121 partial	36.480	1.608	36.948	33.120	38.633	35.330	37.630	4.407
EfficientNet-B0 full	44.642	1.042	45.047	42.846	45.931	43.896	45.388	2.334
EfficientNet-B0 partial	33.323	0.723	33.523	31.779	34.441	32.806	33.841	2.171
DINOv2 full	38.035	1.029	38.278	35.935	39.169	37.299	38.770	2.705
DINOv2 partial	47.112	1.324	46.831	44.872	49.672	46.165	48.060	2.811

TABLE VIII
PER-CLASS FUTURE-YEAR (2026) F1 FOR ALL EIGHT CONDITIONS (MEAN $\pm$ SD OVER 10 SEEDS, %).

Condition	Brown Spot	Healthy	Rice Blast	Rice Tungro
ResNet50 full	45.90 $\pm$ 1.24	80.44 $\pm$ 2.27	10.64 $\pm$ 3.70	38.32 $\pm$ 2.45
ResNet50 partial	38.24 $\pm$ 0.73	67.09 $\pm$ 1.11	11.95 $\pm$ 1.42	39.66 $\pm$ 1.53
DenseNet121 full	45.97 $\pm$ 1.01	82.27 $\pm$ 2.44	11.86 $\pm$ 4.58	39.65 $\pm$ 3.46
DenseNet121 partial	38.96 $\pm$ 3.17	58.90 $\pm$ 5.28	9.77 $\pm$ 3.47	38.30 $\pm$ 1.18
EfficientNet-B0 full	44.10 $\pm$ 1.21	80.99 $\pm$ 2.65	10.89 $\pm$ 2.21	42.59 $\pm$ 4.07
EfficientNet-B0 partial	35.68 $\pm$ 1.21	54.53 $\pm$ 2.11	1.80 $\pm$ 0.37	41.28 $\pm$ 1.19
DINOv2 full	32.67 $\pm$ 2.23	48.39 $\pm$ 2.39	24.10 $\pm$ 3.77	46.98 $\pm$ 2.08
DINOv2 partial	46.41 $\pm$ 1.56	80.61 $\pm$ 1.18	14.78 $\pm$ 4.12	46.65 $\pm$ 2.47

APPENDIX B COLLECTED RICE DISEASE IMAGE DATASETS

Table IX lists the ten publicly available Bangladesh-focused datasets that were combined to build the multi-year corpus described in Section III-A, including source repository, contributing institution, disease coverage, image count, publication date, collection window, and geographic location, where reported.

TABLE IX: Collected rice disease image datasets from Bangladesh. Dates originally stored as Excel serial numbers were converted to calendar dates.

Year	Dataset Name	Source	Institution	Disease Types	Images	Publish Date	Location
2026	Rice Disease Image Dataset	Mendeley	Netrokona University	K def., ear borer, leafroller, stem rot, blight	500	2026-02-24	Netrakona
2026	RiceLeafDiseaseBD (Smartphone)	Mendeley	Independent University	Blast, brown spot, smut, tungro, sheath blight	9,769	2026-03-17	Bangladesh
2025	RiceLeafBD (Real-Field)	Mendeley	Daffodil Int. University	Bact. blight, brown spot, tungro	1,555	2025-11-07	Sylhet, Dhaka
2025	Bangladeshi Crops Disease (BCDD)	Kaggle	UIU + Others	Bact. blight, brown spot, leaf blast	8,992	2025-11-30	N/A
2024	Paddy Leaf Disease	Mendeley	N/A	Blight, brown spot, blast, tungro	N/A	2024-05-21	N/A
2024	Rice Leaf & Pest (BRRI)	Mendeley	Daffodil Int. University	Tungro, stripes, blast, scald, pest	2,769	2024-11-12	Gazipur, Dhaka
2024	Rice Leaf and Crop Disease	Mendeley	Daffodil Int. University	Leaf blight, blast, tungro	2,508	2024-11-20	N/A
2023	Rice Bacterial & Fungal Disease	Mendeley	Khwaja Yunus Ali Univ.	Blight, brown spot, scald, hispa, sheath, blast	1,701	2023-11-28	Sirajganj, Pabna
2022	Paddy Leaf Disease	Kaggle	N/A	Brown spot, healthy, blast, blight, smut	1,115	2022-01-31	N/A
2021	Dhan-Shomadhan Rice Disease	Mendeley	Daffodil Int. University	Brown spot, scald, blast, tungro, sheath blight	1,106	2021-04-06	N/A